

CircuitFlow

Team Manual

PHASE - 3

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CircuitFlow: Smart Water Management — *Drop by Drop*

Project Overview

CircuitFlow is an intelligent domestic water-management framework conceived to mitigate household water wastage through **real-time sensing, automated regulation, and predictive data analytics**. The system continuously monitors water flow at household taps, identifies anomalous or excessive usage, and initiates corrective interventions through hardware-based actuation. In parallel, it analyses historical consumption data to model user behaviour and forecast future usage patterns.

The overarching objective of the project is to foster **sustainable water-use practices** by combining low-power embedded hardware with localized analytics, thereby minimizing dependence on cloud infrastructure while preserving user privacy and system reliability.

Team Composition and Roles

The project is collaboratively developed by **Team Smartopia**, with well-defined functional responsibilities:

- **Hardware Manager – Suhas V J**
  (suhasvj.et25@rvce.edu.in)
  Oversees hardware selection, circuit design, component integration, and physical assembly, including Arduino interfacing.
- **Embedded Systems Manager – Ashi Chauhan**
  (ashichauhan.ec25@rvce.edu.in)
  Responsible for embedded software development, sensor integration, and seamless coordination between hardware and firmware.
- **Data Analyst – Disha G N**
  (gndisha.cs25@rvce.edu.in)
  Manages data acquisition, preprocessing, visualization, and analytical interpretation, along with peer-to-peer data communication.

- **Data Analyst and Team Lead – Gautam Garani**
  (gautamgarani.cs25@rvce.edu.in)
  Leads project execution, supervises analytical modelling, predictive algorithms, data extraction pipelines, and P2P networking.

Project Deliverables

The project aims to produce the following tangible outcomes:

1. A **fully functional prototype** capable of detecting, analyzing, and regulating domestic water flow.
2. A **demonstrable, reproducible, and operational hardware model**, suitable for real-world validation.
3. Comprehensive **data analytics outputs**, including live usage trends, and predictive consumption models.

Project Context and Motivation

The proposed system represents a **holistic domestic water-efficiency solution**, integrating hardware sensing, automated control, and localized data intelligence. The framework consists of:

- A low-power water-flow sensor
- An automated flow-control valve (actuator)
- Auditory and visual alert mechanisms
- A real-time local analytics dashboard
- Peer-to-peer communication without third-party cloud reliance

The sensor records volumetric flow through household taps and transmits this data to a local processing unit. The system subsequently analyses usage behaviour, identifies inefficiencies, and predicts future consumption. Design decisions prioritize **sustainability metrics**, including water-saving potential, energy efficiency, system cost, and scalability.

Existing Solutions and Their Limitations

While several contemporary water-conservation systems exist, they suffer from critical limitations:

- Inability to **adapt to long-term behavioral patterns** or accommodate multiple users.
- Heavy reliance on **internet connectivity and cloud services**, compromising data privacy and increasing operational complexity.
- Dependence on **single-mode sensing techniques**, such as acoustic leak detection, which are prone to inaccuracies in non-controlled environments.
- Bulky designs and **poor cost-to-performance ratios**.
- Lack of temporal optimization, resulting in inefficient power usage and missed peak-wastage periods.

These deficiencies highlight the necessity for a **self-contained, adaptive, and energy-efficient** alternative.

Water Scarcity Crisis and Consumption Trends

Global and National Water Stress

Empirical studies conducted in Indian metropolitan regions indicate average water consumption rates of **183 litres per person per day in standalone households** and **215 litres per person per day in apartment complexes**. Projections suggest that by 2025, nearly **67% of the global population** will experience moderate to severe water stress, with half facing direct supply constraints.

Factors Affecting Domestic Water Consumption

Urban domestic water usage is influenced by multiple parameters, including:

1. Climatic and seasonal variations
2. Types and efficiencies of installed water-use components

3. Individual and collective user behavior

Residential buildings constitute the largest share of urban water consumption, making them critical targets for conservation strategies.

Household Water-Use Components

Water consumption can be disaggregated into micro-components:

- **Bathing and showering** account for approximately **55%** of household water use.
- **Clothes washing** contributes nearly **20%** of per-capita consumption.

Optimising these components through monitoring and control offers substantial conservation potential.

Problem Statement

Rapid population growth has significantly strained global freshwater resources. In India alone, an average household loses approximately **45,500 litres of water annually** due to unattended tap leaks and unregulated flow. Existing solutions largely provide cumulative consumption data without enabling **real-time intervention or intelligent feedback**.

The absence of affordable, automated, and user-centric control mechanisms leads to excessive water loss, particularly during peak usage periods. This project seeks to address these challenges by developing a **smart, adaptive water-usage system** that detects abnormal flow, initiates corrective action, and promotes sustainable behavioral change.

Problem Description and Approach

Traditional plumbing systems lack autonomous regulation and rely solely on user attentiveness. Situations such as unattended taps, minor pipe leaks, or prolonged usage during cleaning activities often go unnoticed, resulting in significant cumulative losses.

The proposed system introduces a **hardware-centric monitoring and regulation solution** that:

- Continuously measures water flow rates
- Detects abnormal or prolonged usage
- Triggers auditory alerts and automated shut-off mechanisms
- Preserves user autonomy during legitimate high-usage events

Rather than enforcing rigid limits, the system adapts warnings based on **historical usage trends**, thereby aligning intervention thresholds with individual household behaviour.

9. System Architecture

9.1 Hardware Components

The system comprises the following core components:

- **Arduino UNO Microcontroller Unit (MCU)**
- USB-B power supply and data interface
- **YF-S401 water-flow sensor**
- **HW201 infrared obstacle detection sensor**
- Proportional solenoid valve (actuator)
- LEDs, resistors, buzzer, and jumper wires

The Arduino operates at **5V**, with **32 KB flash memory** and a **16 MHz clock speed**. The YF-S401 sensor measures flow rates between **0.3–3 L/min** with ±5% accuracy, ensuring safe and reliable operation. The HW201 IR sensor detects hand presence within a configurable range, enabling touchless activation.

9.2 Data and Control Flow

1. Power and firmware are supplied via USB-B.
2. The IR sensor detects the presence of a user's hand.
3. Upon detection, the MCU reads flow data from the water-flow sensor.
4. The solenoid valve opens, allowing water to flow.

5. If the hand is withdrawn without manual shut off, the system gradually restricts flow before fully closing the valve.
6. An auditory alert signals valve closure.
7. Flow metrics are logged and extracted for analysis.
8. LED indicators visually display the flow state (RED: no flow, GREEN: active flow).

Software Architecture and Functional Description

1. Introduction

The main program implements a **real-time water flow monitoring and prediction system** using an Arduino sensor and a Flask-based web application. The system continuously collects water flow data, calculates daily water consumption, predicts future usage using a neural network model, and presents both live and historical data through a web interface.

2. System Objectives

The main objectives of the system are:

- To monitor **real-time water flow** from an Arduino sensor
- To compute **total daily water consumption**
- To store historical water usage data for analysis
- To predict future water consumption using machine learning
- To display live, daily, and overall reports via a web dashboard

3. Time-Based Access Control

The system defines a restricted time window during which certain analytical pages are accessible:

- Start Time: 08:00 AM
- End Time: 10:00 AM

If a user attempts to access the daily analysis page outside this window, an *Unavailable* page is displayed. This ensures controlled access to daily summary data.

4. Arduino Data Acquisition Module

The Arduino is connected to the system through a serial interface. The program continuously:

- Reads water flow values sent by the Arduino
- Converts incoming data into numerical form
- Updates the latest flow value
- Calculates the total volume of water consumed throughout the day

To prevent delays or system blocking, the Arduino data reader runs in a **separate background thread**.

5. Daily Water Usage Logging

At a fixed time the system:

- Records the total water consumed for the day
- Appends this value to a CSV file (`flow_data.csv`)
- Ensures the data is stored only once per day

This CSV file serves as the historical dataset for predictions and reporting.

6. Machine Learning-Based Prediction Module

The system includes a custom-built **neural network**, implemented using NumPy without external machine learning libraries.

Key characteristics:

- Uses the last **four days of water usage data** as input
- Predicts the next day's water consumption
- Applies data normalization for improved training accuracy
- Trains using multiple iterations to minimize prediction error

This approach allows the system to identify trends in water usage and generate future consumption estimates.

7. Daily Prediction Scheduler

A scheduling mechanism triggers the prediction process automatically every day at **a scheduled time.** During this process:

- The trained neural network generates a new prediction
- The predicted value is saved to `daily_predictions.csv`
- The latest prediction is updated for display on the web interface

This operation runs independently in the background.

Web Application (Flask Framework)

The Flask framework is used to create a web-based dashboard with multiple routes:

Home Page

Provides the main entry point to the system.

Live Monitoring Page

Displays real-time water flow data received directly from the Arduino sensor.

Live Data Streaming

The system uses **Server-Sent Events (SSE)** to continuously stream live sensor data to the browser without requiring page refreshes.

Daily Analysis Page

Displays:

- Total water consumed for the current day
- Predicted water usage for the next day
  Access to this page is restricted to the defined time window.

Overall Report Page

Analyzes historical data and presents:

- Average water usage
- Median water usage
- Most frequent usage value (mode)

9. Multithreading and Background Execution

The system runs multiple tasks simultaneously using threads:

- Arduino data acquisition
- Daily prediction scheduling
- Web server execution

Thread locks are used to ensure data consistency when multiple processes access shared variables.